



Socio-demographic patterns of instability

Essays on earnings and income volatility in Belgium

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1. Introduction

2. Structure of the dissertation

- ▶ Research questions
- ▶ Data

3. Empirical findings

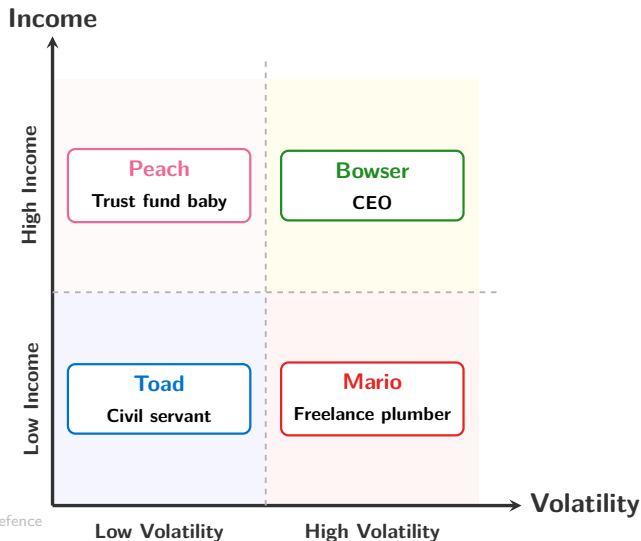
- ▶ Chapter 2 (Earnings inequality, gender and education)
- ▶ Chapter 3 (Migrant-native gap in earnings volatility)
- ▶ Chapter 4 (Social policy design during crises)

4. Conclusion

- ▶ Main contributions
- ▶ Policy implications
- ▶ Limitations and future research

Introduction

What is income volatility?



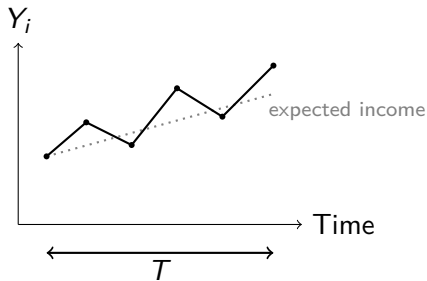
(1) Across-persons / “period-pair”

Dispersion of changes or shocks in income across individuals between two points in time.

	Year 1		Year 2
Worker A	€30k	→	€50k (+€20k)
Worker B	€40k	→	€35k (-€5k)
Worker C	€25k	→	€70k (+€45k)

(2) Within-person / “multi-period”

Changes over a specified time window for a given individual; the degree of deviation from their expected level of income.



Milton Friedman's (1957) PIH:

$$\text{Income} = \text{Permanent Income} + \text{Transitory Income}$$

- ▶ **Permanent Income** = long-run average income
 - Due to time-invariant individual-specific characteristics (e.g. innate ability, skills, schooling)
 - **Persistent** gains in lifetime earnings
- ▶ **Transitory Income** = temporary deviations from permanent income (i.e. earnings profile)
- ▶ **PIH** → consumption only reacts to permanent income
 - Individuals smooth consumption by saving or borrowing in response to temporary income shocks

Introduction

Income dynamics and inequality



$$\text{Income} = \text{Permanent Income} + \text{Transitory Income}$$

$$\text{Var}(\text{Income}) = \text{Var}(\text{Permanent}) + \text{Var}(\text{Transitory})$$

$$\text{Total Inequality} = \text{Permanent Inequality} + \text{Income Instability}$$

Different drivers = different policy implications

- ▶ **Permanent** : shifts in labour demand (e.g. skill-biased technological change, changes in returns to education) → **human capital investment**
- ▶ **Transitory** : labour market (in)stability (e.g. flexible employment, incidence of job-switching) → **income stabilisation**

Introduction

Why study volatility?

Potential consequences¹:



Physical health



Mental health



Intergenerational effects

¹See e.g., Dahl et al. (2014) on food insecurity; Ziff et al. (2025) on life expectancy; Schneider and Harknett (2019) on stress; Hill et al. (2013) on children's wellbeing; Shore (2011) on intergenerational transmission.

Introduction

Why study volatility **now**?



- ▶ Concerns over the erosion of the **standard employment relationship**
- ▶ Globalisation and technological change demand (numerical) **flexibility** (Kalleberg, 2009)
→ (partial) **deregulation** of labour markets and employment protection legislation (EPL)
- ▶ The global rise of **non-standard work** (Zwysen, 2025)



Introduction

Why study volatility **now**?



- ▶ Adaptation of the **flexicurity** model
 - Advocated by the European Commission in the 2007 European Employment Strategy
 - Critiques: “[...] *flexicurity is essentially the sugar coating on a bitter liberalization pill*” (Mailand, 2010)
- ▶ Weakening of social protection and welfare systems in some countries (Avram et al., 2022; Garcia-Medina and Wen, 2018; Hardy, 2016; Hacker, 2019)
- ▶ **Polycrisis**: COVID-19 pandemic, energy crisis, inflation, geopolitical instability, climate change



Introduction

Why study volatility in Belgium ?



- ▶ Low levels of wage inequality relative to the rest of EU (Assal et al., 2022), but with a **segmented labour market**
 - Especially along the lines of education and migration background (Gonne, 2022)
- ▶ **Non-standard work** is also on the rise in BE (Nautet and Piton, 2019)
 - Incidence of part-time employment is above EU average, especially among women
- ▶ **Not much is known** about earnings (and income) volatility in Belgium²
→ the **main contribution** of this dissertation is filling this gap

²Some earlier studies in a cross-national setting: Sologon and O'Donoghue (2009); Ayllón and Ramos (2019); Venn (2011)

This dissertation

Research questions

1. How has earnings volatility **evolved in Belgium**, and to what extent do aggregate inequality trends conceal shifts between permanent and transitory components?

Chapter 2

2. How do volatility levels and trends differ across different **socio-demographics**?

Chapter 2

Chapter 3

3. How sensitive are these differences and trends to **measurement choices**?

Chapter 3

4. How does **policy design** shape the effectiveness of income-protection policies in mitigating shocks, particularly during **crises**?

Chapter 4

Two data sources:

1. Linked administrative registers

Chapter 2

Chapter 3

- Novel longitudinal dataset from 2005 to 2021
- Combines tax returns (IPCAL) with census data and the demographic register, DEMOBEL (10% random sample of tax-filers)

2. Nowcasted and microsimulated survey data

Chapter 4

- Belgian EU-SILC data in 2019
 - Updated and nowcasted to 2020
 - Labour market transitions (non-parametrically) assigned using administrative data based on socio-demographic characteristics
- Microsimulation model (EUROMOD) to simulate changes to social policies during the COVID-19 pandemic at an intra-year level

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CH.2: The shift from persistent inequality to earnings instability in Belgium

A decomposition by gender and education



► Method:

- Error-components model with Generalised Method of Moments (GMM) (i.e., variance decomposition into permanent and transitory components)
- Biewen decomposition of permanent inequality

► Main findings:

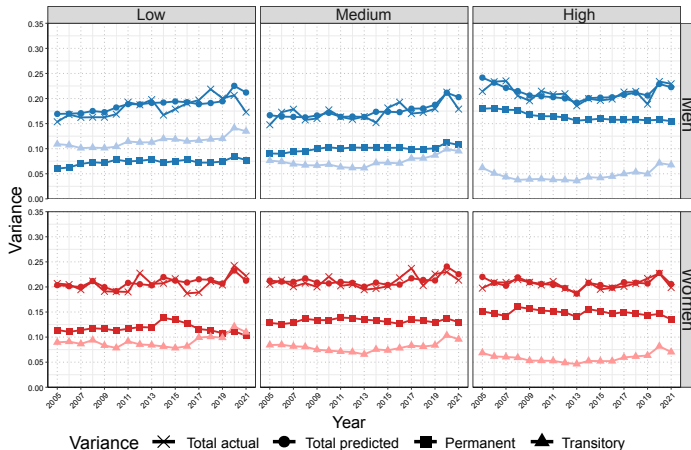
- Earnings inequality is quite stable with a slight increase after 2013
- Underlying: permanent inequality decreased, while earnings instability increased after 2013
- Education gradient in terms of levels of earnings instability and persistence of transitory shocks

The less educated have more volatile earnings

And younger cohorts are also more volatile



Figure 1: Variance decomposition, mid-career worker by gender and education



Decomposing overall permanent inequality

Biewen decomposition



- ▶ Attributing trends to changes in the share of each gender-education subgroup and changes in the level of permanent inequality within each subgroup
- ▶ **Main findings:**
 - Increasing educational attainment has a disequalising effect
 - But this was offset by decreases within subgroups and also in inequality between genders



CH.3: Lower pay and less stable?

Understanding the migrant-native gap in earnings volatility



Two parts: (I) comparing measurement approaches and;
(II) understanding the factors that drive the migrant-native gap in earnings

► Part I Method:

- Parametric (i.e., CH.2) and non-parametric “summary” approaches³ to measuring volatility

► Part II Method: (decompositions of summary volatility)

- Labour market transitions → Variance decomposition
- Socio-demographic drivers → Recentred Influence Function-Oaxaca Blinder (RIF-OB) decomposition (Firpo et al., 2018)
 - “Composition” and “structure” effects

³i.e., dispersion of arc percentage changes and log changes, coefficient of variation

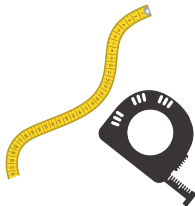
CH.3: Lower pay and less stable?

Part I: Measurement



► Main findings:

- Trends differ across approaches after 2013
 - Summary volatility declined but earnings instability (parametric method) increased
→ this increase was only observed for natives and second-generation migrants
- Consistent across approaches: significant differences in volatility levels among migrant groups
 - First-generation non-EU migrants have the highest volatility levels and persistence of transitory shocks
- Also consistent: convergence of volatility levels between natives and migrants
 - Driven by decline in volatility among migrants



CH.3: Lower pay and less stable?

Part II: Explaining the trends & drivers of the migrant-native gap



► **Main findings:** *Labour market transitions*

- Overall declines in SV → declines in volatility within employment
 - Equally for those who remain employed and those transitioning into and out of work
 - BUT employment shares decreased while shares *remaining out of work* increased
- Self-employment volatility: transitions into and out of self-employment drive earnings volatility for first-generation migrants

► **Main findings:** *Socio-demographic drivers*

- Gap is mostly **unexplained** (“structural”)
 - **Origin-based** differences: more than 90% unexplained
- **Composition:** 🎓 **education** and 🌐 **region** are major drivers
- **Structure:** 👤 **gender** and 🏠 **household** characteristics are major drivers
- **Unobserved** structural factors (e.g., discrimination, social networks) are increasingly important

CH.3: Lower pay and less stable?

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CH.4: Balancing timeliness, efficiency and effectiveness

An assessment of targeted social policies in challenging times



► Context:

- COVID-19 pandemic and the Belgian policy response
- Extensions to existing measures
 - Temporary unemployment and the bridging right for the self-employment
- Discretionary measures in 2020
 - One-off bonuses for social assistance, utility and rental premiums in different regions
- Framework: targeting design (Clasen and Clegg, 2007)
 - Categorical-based
 - Circumstance-based – direct means-test or proxied by “passporting”



CH.4: Balancing timeliness, efficiency and effectiveness



► Method:

- **Timeliness**: decomposing the coefficient of variation (intra-year income volatility)
- **Efficiency**: the distributional characteristic index (DCI)
- (Poverty Reduction) **Effectiveness**: Beckerman ratios

► Main findings:

- Pre-existing, large-scale benefits were the most **effective** and **timely**
- Circumstance-targeted benefits were generally the most **redistributively efficient**
 - BUT doesn't guarantee substantial **poverty reduction**
- No clear trade-off between redistributive efficiency and timeliness
 - **Passporting** → not necessarily timely

Conclusion

Main contributions



- ▶ **First analyses** on long-run inequality and earnings volatility trends in Belgium
- ▶ Exploring **heterogeneity** in volatility levels and trends across socio-demographic groups
 - Gender, education and migration background
- ▶ How **demographic shifts** impact overall permanent inequality → Biewen decomposition
- ▶ **Factors explaining differences** in volatility between subgroups → RIF-OB decomposition
- ▶ **Timeliness** as an additional dimension to be evaluated in **social policy design**

- ▶ Importance of **human capital** for stability in the labour market
 - Upskilling and reskilling as policies to improve labour market outcomes
 - Early-childhood interventions (re: tracking/streaming)
- ▶ **Migrants** not only have lower pay but also **less predictable** trajectories
 - Reducing barriers to social and economic integration
 - Social networks may be important (but we don't know much)
- ▶ Importance of **social protection** and income stabilisation policies (COVID-19)
 - Especially in light of recent austerity measures...
 - To target or not to target?

Conclusion

Limitations and future research



- ▶ Inability to make **causal** claims about the drivers of volatility trends and gaps
 - Impact of related employment policies (and reforms) in Belgium
- ▶ **Annual** nature of tax returns and data limitations for **other income**
 - Volatility in hours, wages and disposable incomes, social protection, family dynamics, consumption volatility
- ▶ Volatility → **precarity**? Not always
 - Changing preferences for working arrangements and economic insecurity



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Thank you for attending and listening!
It's now time for questions from the jury (and audience).

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Appendix

Contents



1. ▶ Industrial change in Belgium
2. ▶ Labour force participation (LFP) rates
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13. ▶ Non-standard work in Belgium II: flexi-jobs
14. ▶ Non-standard work in Belgium III: bogus self-employed

Appendix

Industrial change in Belgium



Table A1: Gender and skill distribution in growing and declining sectors and occupations

Category	Sectors		Occupations	
	Growing	Declining	Growing	Declining
Gender				
Men	32	77	45	94
Women	68	23	55	6
Education Level				
Low	15	24	4	28
Medium	32	51	22	62
High	53	25	73	10

Source: Deschacht and Detilleux (2023), based on NACE- and ISCO-harmonised EAK (i.e., Belgian Labour Force Survey).

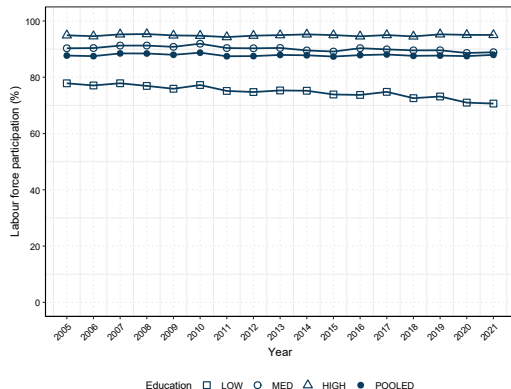
Note: Sector shares are based on data from 2010–2020. Occupation shares are based on data from 2013–2020. Growing sectors = e.g., cleaning, security, hospitals and computer programming; Declining sectors = e.g., manufacturing, agriculture, postal services. Growing occupations = e.g., business administration, health professionals, social workers and ICT workers; Declining occupations = blue-collar occupations in trade and manufacturing.

Appendix

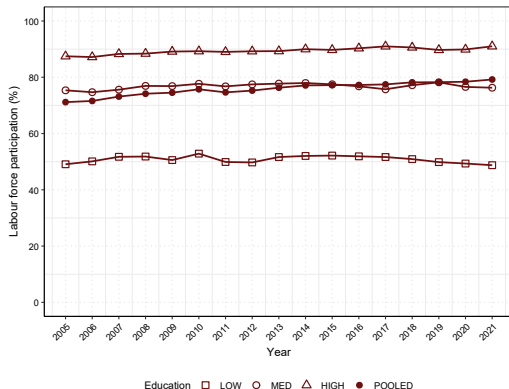
Labour force participation (LFP) rates

Figure A1: LFP by gender and education level, 2005-2021

(a) Males



(b) Females



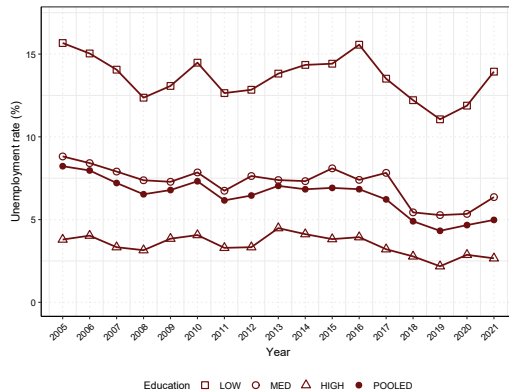
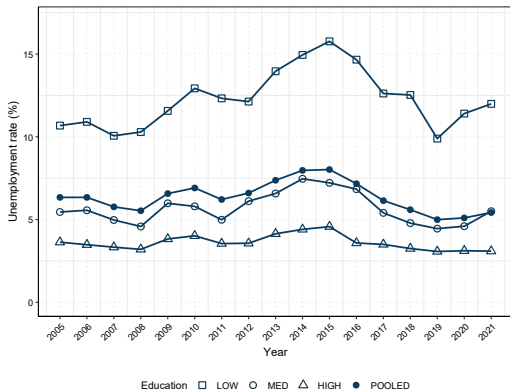
Source: Own calculations, based on Labour Force Survey (LFS).

Note: Share of active labour force (i.e., employed and unemployed persons) in each subgroup, calculated among individuals of working age (i.e., between the ages of 25 and 59).

Appendix

Unemployment rates

Figure A2: Unemployment rates by gender and education level, 2005-2021
(a) Males (b) Females



Source: Own calculations, based on Labour Force Survey (LFS).

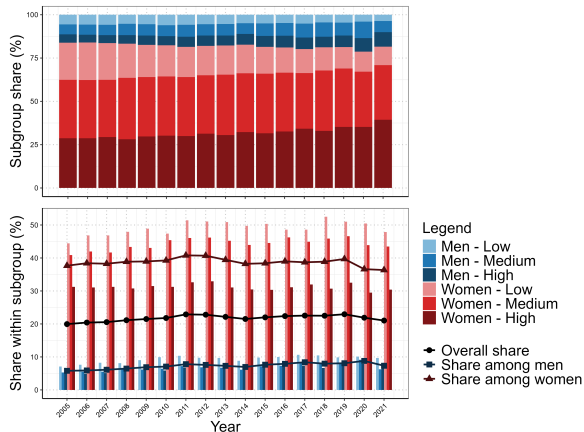
Note: Unemployment as a share of the labour force in each subgroup, calculated among individuals of working age (i.e., between the ages of 25 and 59).

Appendix

Trends in part-time work



Figure A3: Gender and education distribution of part-time work, 2005-2021



Source: Own calculations, based on Labour Force Survey (LFS) data.

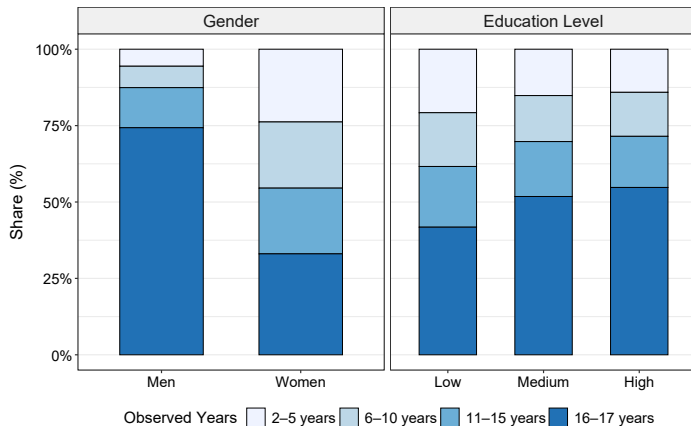
Note: Working age (25-59) population.

Appendix

Attrition in Chapter 2 (broad subgroups)



Figure A4: Shares per range of positive earnings years by gender and education level



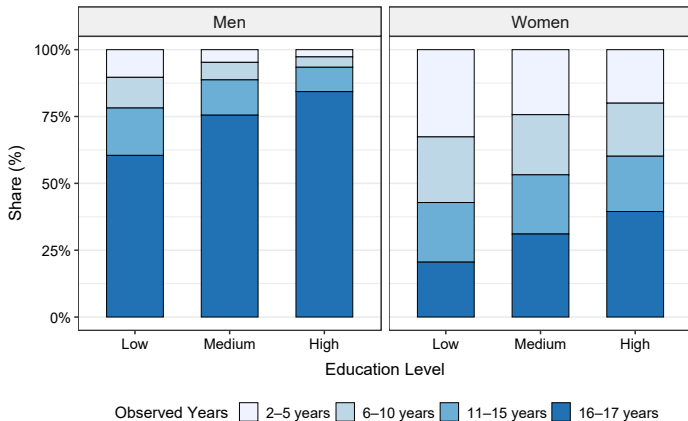
Note: Share of individuals in each bin relative to total number of individuals in each subgroup with minimum two years of positive earnings.

Appendix

Attrition in Chapter 2 (gender-education subgroups)



Figure A5: Shares per range of positive earnings years by gender-education

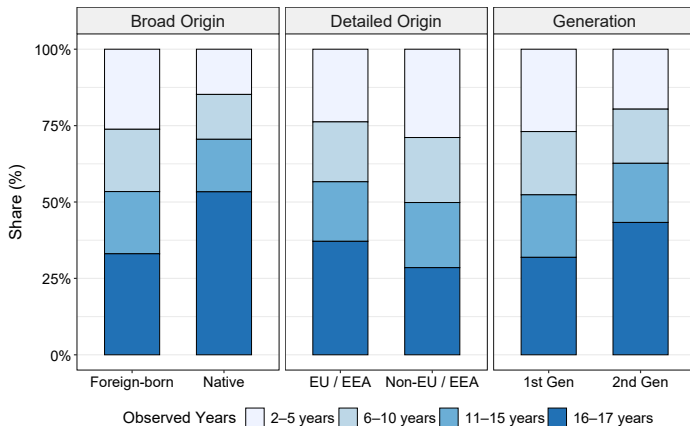


Note: Share of individuals in each bin relative to total number of individuals in each subgroup with minimum two years of positive earnings.

Appendix

Attrition in Chapter 3

Figure A6: Shares per range of positive earnings years by migration background

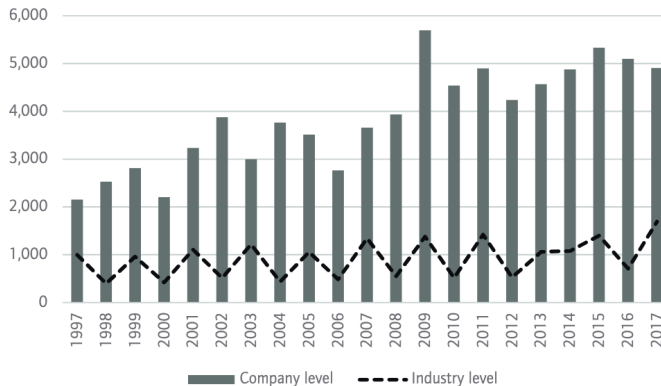


Note: Share of individuals in each bin relative to total number of individuals in each subgroup with minimum two years of positive earnings.

Appendix

Collective bargaining trends in Belgium

Figure A7: Collective agreements at the company and industry levels, 1997–2017



Source: Vandaele (2019).

Note: Collective bargaining coverage has remained high in Belgium (96%) and trade union density remained stable (around 55%) relative to deunionisation trends more broadly in the EU. However, company-level collective agreements (e.g., on non-recurring bonuses) have grown.

Temporary fixed-term contracts

- ▶ Belgian law (Employment Contracts Act of 1978) explicitly prohibits rolling fixed-term contracts as a standard practice
 - Maximum duration of 24 months (maximum of 4 consecutive contracts)
→ > 24m? contract is converted to permanent
 - 2023 reform: abolished **loophole** allowing successive fixed-term and replacement contracts

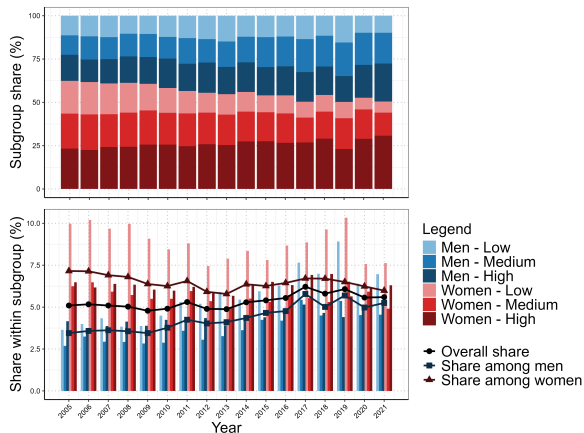
Temporary agency work (TAW)

- ▶ For replacing permanent employees, temporary increases in workload, “probation” placements, performing extraordinary tasks, activation of long-term unemployed
 - Can generally be successive within the above activities
- ▶ EU directive regarding equal treatment of TAW and permanent workers
 - Belgian adaptation goes above and beyond but **in practice TAW does not always receive equal treatment** w.r.t. fringe benefits (Bosmans et al., 2022)

Appendix

Trends in temporary contracts

Figure A8: Gender and education distribution of temporary contracts, 2005-2021



Source: Own calculations, based on Labour Force Survey (LFS) data.

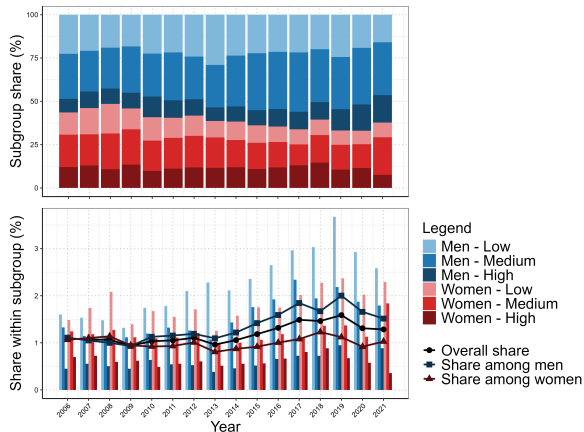
Note: Working age (25-59) population.

Appendix

Trends in TAW



Figure A9: Gender and education distribution of TAW, 2005-2021



Source: Own calculations, based on Labour Force Survey (LFS) data.

Note: Working age (25-59) population.

Flexi-jobs (since December 2015)

- ▶ Workers employed 4/5 FTE can be employed in a second job with tax and social security exemptions on this second income
- ▶ Started in the hospitality sector and has been expanded to other sectors
- ▶ Critiques:
 - **Rules can be circumvented:**
 - “[...] it is perfectly possible for a worker to be **solely employed in a flexi-job**. This is because the condition of main activity for another employer is not checked in real time [...]” (Dermine and Mechelynck, 2022);
 - Businesses operating across multiple legal entities
 - A form of **“zero-hours” contracts**: unpredictable work schedules
 - Replacing permanent workers split across several flexi-jobbers

Appendix

Non-standard work in Belgium III: bogus self-employed



Platform work (i.e., gig economy)

- ▶ 2016 law on special ad-hoc “sharing economy” status included some tax reductions on income up to €5000; inadequate especially w.r.t. social protection (Wattecamps, 2021)
- ▶ Frequent changes to legislation created uncertainty for workers with multiple regimes (Lenaerts and Gillis, 2025)
 - 2018 law: higher income ceiling fully tax- and SSC-exempt → abolished in 2020
 - 2022 Labour Deal: platform workers are considered mostly as employees
 - 2026: work accident insurance must be provided by platforms
 - 2016 still applies to most in food delivery

Freelancing

- ▶ Cheaper for companies to hire (taxes, SSC, fringe benefits); no social protection for the actual employee
- ▶ National Office Social Security (RSZ) does audits to determine genuine self-employment vs. disguised employment
 - If misclassified, company must pay back social security contributions and fines

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